MIL-P-37263A
30 April 1976
SUPERSEDING
MIL-P-37263
3 July 1975

MILITARY SPECIFICATION

POLYVINYL ALCOHOL OPHTHALMIC SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Polyvinyl Alcohol Ophthalmic Solution (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a sterile ocular lubricant containing the following formula:

Polyvinyl Alcohol	1.40%
Chlorobutanol	0.5%
Sodium Chloride	0.78%
Purified Water	qs to 100% W/V

3.2 Assay. The ophthalmic solution shall assay to contain not less than 1.20 percent and not more than 1.60 percent Polyvinyl Alcohol when assayed as specified in 4.5.1.

3.3 Identity. The polyvinyl alcohol in the ophthalmic solution shall comply with the identity test specified in 4.5.2.

3.4 Ingredients.

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3.4.1 Polyvinyl alcohol, chlorobutanol, sodium chloride and purified water. The polyvinyl alcohol, chlorobutanol, sodium chloride and purified water used in the manufacture of the ophthalmic solution shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.4.2 Other ingredients. All other ingredients used in the manufacture of the product shall be of U.S.P. or N.F. quality, or, if not included in either of these compendia, the ingredients shall be suitable for use in this preparation.

3.5 Sterility. The ophthalmic solution shall be sterile when tested in accordance with the U.S.P. sterility test method.

3.6 Expiration dating period. The expiration dating period shall be not less than 48 months.

3.6.1 Delivery. Not more than 4 months of the expiration dating period shall have elapsed at the time of delivery of the product to the Government.

3.7 Workmanship. Workmanship shall be first class throughout. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of any inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purpose of this specification, a lot, batch, or control, is that single, uniform, and homogeneous quantity of ophthalmic solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final container.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Raw Material and End Item Testing

Characteristic	Requirement paragraph	Inspection level 1/	Minimum Sample
Assay	3.2	S-2	5
Identification	3.3	S-2	5
Ingredients	3.4	2/	-

NOTE: For all tests including sterility test, the acceptance number shall be zero.

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform test indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

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4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects

Categories	Defects 1/
Major	
101	The ophthalmic dropper container does not function properly.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5.1 Assay for polyvinyl alcohol in the finished solution.

Reagents.

1. Boric Acid Solution.
   Dissolve 32 grams of Boric Acid in purified water and q.s. to 1000 ml.

2. Iodine/Potassium Iodide Solution.
   Weigh about 12.7 grams of iodine and 25 grams of potassium iodide into a 250 ml beaker, add about 50 ml of purified water and dissolve. Transfer the solution into a 1000 ml volumetric flask and q.s.with purified water. Store this solution in an amber bottle.

Standard preparation.

Weigh accurately about 70 mg of standard polyvinyl alcohol into a 100 ml volumetric flask; add hot purified water, dissolve by stirring continuously with a magnetic stirrer/heater. Cool, and q.s. with purified water.

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Procedure.

NOTE: The following solutions and readings shall be conducted between 20°-23°C.

Pipet 5.00 ml of sample into a 100 ml volumetric flask, bring to volume with purified water, and mix well. Accurately transfer 1.00 ml of this solution to another 50 ml volumetric flask. Into a second 50 ml volumetric flask pipet 1.00 ml of the standard preparation. Use a third 50 ml volumetric flask for a reagent blank.

Add to all flasks 25 ml of purified water, 15 ml of the boric acid solution and 0.50 ml of the iodine-potassium iodide solution. Bring to volume with purified water. Mix well and allow the solution to stand for 5 minutes.

Concomitantly determine the absorbances of the standard and the sample solutions at 660 mu on a suitable spectrophotometer against the reagent blank.

Calculate the percent purity as follows:

$$\text{Calculation: } \frac{Au}{As} \times \frac{Cs}{Cu} \times 100$$

where Au = absorbance of the sample solution.
As = absorbance of the standard solution.
Cs = concentration of standard solution (mg/ml)
Cu = concentration of sample solution (mg/ml)

4.5.2 Identification of polyvinyl alcohol in the finished solution.

Reagents.

1. 0.1M Boric Acid.
   Dissolve 0.6 grams of boric acid in enough purified water to make 100 ml.

2. Iodine-Potassium Iodide Solution
   Dissolve 0.254 grams of iodine and 21.4 grams potassium iodide in 25 ml of purified water, then add enough purified water to make 1000 ml.

Procedure.

Pipet 1 ml of the sample solution into a test tube, add 1 ml of the 0.1M boric acid and mix. Then add 1 ml of the potassium tri-iodide solution and mix. A red color should develop.

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5.6 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Labeling inspection. Sampling shall be in accordance with MI MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 (see 6.3), and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 15 ml, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available plastic dropper bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

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5.2.2.3 Intermediate package. Twelve units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTE: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Labeling adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container.
(5) If a unit package is supplied, the expiration date may appear on the unit package in lieu of the immediate container.

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5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with paragraph 5.5.3 of PPP-C-186. Type II Shelf Life Marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type II Shelf Life Marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-961-7486	POLYVINYL ALCOHOL OPHTHALMIC SOLUTION, 1.4%, 15 ml

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1572

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-37263A POLYVINYL ALCOHOL OPHTHALMIC SOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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